

Unit 4: Autonomous Equations, Equilibrium Solutions, and Stability

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Master Unit Reference Guide: Cheat Sheet, Problems, and Solutions

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Unit 4 at a Glance: Core Sub-Topics

This unit studies first-order equations *qualitatively*: when the right side depends on y alone, the long-run behavior can be read off without ever solving. It then closes with the Bernoulli family, the one nonlinear form that a clever substitution turns linear.

- **Equilibrium Solutions and Stability** — constant solutions $f(c) = 0$, and the sign of $f'(c)$ that makes them attract or repel.
- **Autonomous Equations and the Phase Line** — $y' = f(y)$ as a one-dimensional flow, with arrows set by the sign of f .
- **The Logistic Equation** — $P' = rP(1 - P/K)$, its two equilibria, and the carrying capacity K .
- **Bernoulli Differential Equations** — the form $y' + P(x)y = Q(x)y^n$ and the linearizing substitution $v = y^{1-n}$.
- **Solving Bernoulli Equations** — the full divide, substitute, solve-linear, and back-substitute workflow.
- **Applications of Bernoulli Equations** — population models and the fluid-energy form along a streamline.

Part I — Condensed Cheat Sheet

4.1 Equilibrium Solutions and Stability

An *equilibrium* (steady state) of $\frac{dy}{dt} = f(y)$ is a constant solution $y = c$. A constant has zero derivative, so the rate must vanish there: $f(c) = 0$. Whether nearby solutions return to c or flee from it is decided by the sign of the slope $f'(c)$.

Equilibria and the Linearized Stability Test

$y = c$ is an equilibrium of $\frac{dy}{dt} = f(y)$ exactly when

$$f(c) = 0.$$

Classify it by the sign of $f'(c)$:

$$\begin{array}{ll} f'(c) < 0 & \text{stable (a **sink**) — solutions return} \\ f'(c) > 0 & \text{unstable (a **source**) — solutions flee} \\ f'(c) = 0 & \text{test fails; check the sign of } f \text{ on each side} \end{array}$$

When f touches zero *without changing sign*, the point is **semi-stable**: it attracts from one side and repels from the other.

Worked Example 4.1: Classify the equilibria of $\frac{dy}{dt} = y - y^3$

Step 1. Set $f(y) = 0$ and factor:

$$y - y^3 = y(1 - y^2) = y(1 - y)(1 + y) = 0 \implies y = -1, 0, 1.$$

Step 2. Test each root with $f'(y) = 1 - 3y^2$:

$$f'(0) = 1 > 0 \text{ (unstable),} \quad f'(\pm 1) = -2 < 0 \text{ (stable).}$$

So $y = \pm 1$ are sinks and $y = 0$ is a source.

4.2 Autonomous Equations and the Phase Line

An equation is *autonomous* when its right side depends on y alone, with no explicit t : $\frac{dy}{dt} = f(y)$. Because nothing depends on the clock, any time-shift $y(t - c)$ of a solution is again a solution — solution curves are horizontal translates of one another. All the behavior collapses onto a single *phase line*: a y -axis with arrows.

Reading the Phase Line

Mark the equilibria ($f = 0$) on the y -axis. Between them the sign of f sets the arrow:

$$f(y) > 0 \implies \text{arrow up}, \quad f(y) < 0 \implies \text{arrow down}.$$

arrows in from both sides	stable sink
arrows out from both sides	unstable source
one in, one out	semi-stable

Worked Example 4.2: Phase line for $\frac{dy}{dt} = y^2 - 4$

Step 1. Equilibria where the rate vanishes:

$$y^2 - 4 = 0 \implies y = -2, 2.$$

Step 2. Test the rate sign in each interval:

$$y < -2: (+) \uparrow, \quad -2 < y < 2: (-) \downarrow, \quad y > 2: (+) \uparrow.$$

Arrows converge on -2 (stable) and diverge from 2 (unstable).

4.3 The Logistic Equation

The logistic model corrects exponential growth with a crowding term, so growth slows as the population nears its ceiling. For small P it behaves like exponential growth $P' \approx rP$; near the ceiling the growth shuts off.

The Logistic Equation and Its Equilibria

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right), \quad r > 0.$$

- Equilibria: $P = 0$ (**unstable**) and $P = K$ (**stable**).
- K is the **carrying capacity** — the population level the solution approaches.
- Growth rate is greatest at the midpoint $P = \frac{K}{2}$, where the downward parabola in P peaks.

Common Pitfall: Carrying-Capacity Traps

K is a *population level*, not a growth rate — the intrinsic rate is the constant r . The peak growth sits at $\frac{K}{2}$, not at K (where growth is zero). And do not drop the $P = 0$ equilibrium: with $r > 0$ it is *unstable*, while $P = K$ is the *stable* ceiling — reversing these two is the classic sign error.

Worked Example 4.3: Evaluate the logistic rate

With $r = 0.5$, $K = 100$, find $\frac{dP}{dt}$ at $P = 25$:

$$\frac{dP}{dt} = 0.5 \cdot 25 \left(1 - \frac{25}{100} \right) = 12.5 \cdot 0.75 = 9.375.$$

The population is below $\frac{K}{2} = 50$, so it is still accelerating toward the fastest-growth point.

4.4 Bernoulli Differential Equations

A *Bernoulli equation* looks like a first-order linear equation but carries a power of y on the right. That single nonlinear term is exactly what the substitution $v = y^{1-n}$ removes.

Standard Form and Linearizing Substitution

$$y' + P(x)y = Q(x)y^n.$$

- For $n = 0$ or $n = 1$ the equation is already linear — no substitution needed.
- For any other n , set

$$v = y^{1-n}, \quad \frac{dv}{dx} = (1-n)y^{-n} \frac{dy}{dx},$$

which turns the equation into a **first-order linear** equation in v .

Why $v = y^{1-n}$ Works

Dividing $y' + Py = Qy^n$ by y^n gives $y^{-n}y' + P y^{1-n} = Q$. The group y^{1-n} is exactly v , and $y^{-n}y'$ is $\frac{1}{1-n} v'$, so the whole equation becomes

$$\frac{1}{1-n} v' + P v = Q \implies v' + (1-n)P v = (1-n)Q,$$

a clean linear equation ready for an integrating factor.

4.5 Solving Bernoulli Equations, Worked Examples

The solution is a fixed four-step pipeline: clear the power, substitute, solve the linear equation with an integrating factor, then undo the substitution.

The Bernoulli Solution Algorithm

1. **Divide** every term by y^n .
2. **Substitute** $v = y^{1-n}$, $v' = (1-n)y^{-n}y'$ to get $v' + (1-n)P v = (1-n)Q$.
3. **Solve** the linear equation with $\mu = e^{\int (1-n)P dx}$.
4. **Back-substitute** $v = y^{1-n}$ and solve for y .

Worked Example 4.5: Reduce $y' + y = xy^3$

Step 1. Here $n = 3$, so $v = y^{1-3} = y^{-2}$ and $v' = -2y^{-3}y'$.

$$\text{Divide by } y^3 : \quad y^{-3}y' + y^{-2} = x.$$

Step 2. Substitute, using $y^{-3}y' = -\frac{1}{2}v'$:

$$-\frac{1}{2}v' + v = x \implies v' - 2v = -2x,$$

a first-order linear equation in v ready for $\mu = e^{-2x}$.

4.6 Applications of Bernoulli Equations

The name attaches to two distinct ideas. The Bernoulli *differential equation* models constrained growth and resistive dynamics; the Bernoulli *principle* in fluids is an energy balance along a stream-

line.

The Fluid Bernoulli Equation

Along a streamline, mechanical energy per unit volume is conserved:

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}.$$

The three terms are pressure, kinetic, and potential energy density. With the total fixed, a faster flow ($v \uparrow$) forces a lower pressure ($P \downarrow$); because the kinetic term scales with v^2 , doubling the speed quadruples it.

Common Pitfall: Two Different “Bernoulli” Equations

The Bernoulli *ODE* $y' + P(x)y = Q(x)y^n$ and the fluid Bernoulli *principle* share only a name. Do not import the substitution $v = y^{1-n}$ into a fluids problem, or treat $\frac{1}{2}\rho v^2$ as something to linearize — one is a solution method, the other a conservation law.

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb from locating equilibria, through stability tests and phase lines, into the logistic model, and finish with the Bernoulli substitution $v = y^{1-n}$ and a full solve. Solutions follow in Part III.

1. Find all equilibrium solutions of $\frac{dy}{dt} = y(y - 3)$.
2. Classify the equilibrium $y = 2$ of $\frac{dy}{dt} = y - 2$ using the sign of f' .
3. For $\frac{dy}{dt} = y - y^3$, find every equilibrium and classify each with the linearized test.
4. Classify the equilibrium $y = 1$ of $\frac{dy}{dt} = (y - 1)^2$. Why does the f' test alone not settle it?
5. Build the phase line for $\frac{dy}{dt} = y^2 - 4$: mark the equilibria, set the arrows, and classify each.
6. For the logistic equation $\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$ with $r > 0$, find both equilibria, classify them, and give the population of fastest growth.
7. With $r = 0.5$ and $K = 100$, compute $\frac{dP}{dt}$ at $P = 25$ and state the population at which the growth rate is greatest.
8. For $y' + y = xy^3$, identify n , write the substitution v , and compute $\frac{dv}{dx}$ in terms of y' .
9. Reduce $y' + \frac{1}{x}y = xy^2$ to a first-order *linear* equation in $v = y^{1-n}$ (do not solve it yet).
10. **(Capstone)** Solve the Bernoulli equation $y' + y = y^2$ completely: reduce it with $v = y^{-1}$, solve the linear equation, and back-substitute for y .

Part III — Complete Solutions

Solution 1. Equilibria are the roots of $f(y) = 0$:

$$y(y - 3) = 0 \implies y = 0 \text{ and } y = 3.$$

Solution 2. Here $f(y) = y - 2$, so $f'(y) = 1$:

$$f'(2) = 1 > 0 \implies \text{unstable (a source)}.$$

A positive slope of f at the equilibrium repels nearby solutions.

Solution 3. Factor for the equilibria, then apply $f'(y) = 1 - 3y^2$:

$$y - y^3 = y(1 - y)(1 + y) = 0 \implies y = -1, 0, 1.$$

$$f'(0) = 1 > 0 \text{ (unstable),} \quad f'(\pm 1) = 1 - 3 = -2 < 0 \text{ (stable).}$$

So $y = \pm 1$ are sinks and $y = 0$ is a source.

Solution 4. The only equilibrium is $y = 1$. The linearized test is inconclusive because

$$f'(y) = 2(y - 1) \implies f'(1) = 0.$$

Fall back on the sign of $f(y) = (y - 1)^2$, which is ≥ 0 everywhere and *never* changes sign across $y = 1$:

$$y < 1: f > 0 \text{ (}\uparrow\text{),} \quad y > 1: f > 0 \text{ (}\uparrow\text{)}.$$

Solutions rise toward 1 from below but away from 1 above, so $y = 1$ is **semi-stable**.

Solution 5. Equilibria where the rate vanishes:

$$y^2 - 4 = 0 \implies y = -2, 2.$$

Test the rate sign on each interval:

$$y < -2: (+) \uparrow, \quad -2 < y < 2: (-) \downarrow, \quad y > 2: (+) \uparrow.$$

Arrows converge on -2 (**stable**) and diverge from 2 (**unstable**).

Solution 6. Set the product to zero:

$$rP\left(1 - \frac{P}{K}\right) = 0 \implies P = 0 \text{ and } P = K.$$

Classify with $f'(P) = r\left(1 - \frac{2P}{K}\right)$:

$$f'(0) = r > 0 \text{ (unstable),} \quad f'(K) = r(1 - 2) = -r < 0 \text{ (stable).}$$

The growth rate, a downward parabola in P with roots 0 and K , peaks at the midpoint:

$$P = \frac{K}{2}.$$

Solution 7. Substitute into $rP(1 - \frac{P}{K})$:

$$\frac{dP}{dt} = 0.5 \cdot 25 \left(1 - \frac{25}{100} \right) = 12.5 \cdot 0.75 = 9.375.$$

The greatest growth occurs at $P = \frac{K}{2} = 50$.

Solution 8. The right side is y^3 , so $n = 3$ and the linearizing exponent is $1 - n = -2$:

$$v = y^{1-n} = y^{-2}, \quad \frac{dv}{dx} = (1-n)y^{-n} \frac{dy}{dx} = -2y^{-3} y'.$$

Solution 9. Here $n = 2$, so $v = y^{-1}$ and $v' = -y^{-2}y'$. Divide every term by y^2 :

$$y^{-2}y' + \frac{1}{x}y^{-1} = x.$$

Replace $y^{-1} = v$ and $y^{-2}y' = -v'$:

$$-v' + \frac{1}{x}v = x \implies v' - \frac{1}{x}v = -x,$$

a first-order linear equation in v .

Solution 10. With $n = 2$, set $v = y^{-1}$, $v' = -y^{-2}y'$. Divide $y' + y = y^2$ by y^2 :

$$y^{-2}y' + y^{-1} = 1.$$

Substitute $y^{-1} = v$, $y^{-2}y' = -v'$:

$$-v' + v = 1 \implies v' - v = -1.$$

Integrating factor $\mu = e^{-x}$:

$$(e^{-x}v)' = -e^{-x} \implies e^{-x}v = e^{-x} + C.$$

So $v = 1 + Ce^x$, and back-substituting $v = y^{-1}$:

$$y = \frac{1}{v} = \frac{1}{1 + Ce^x}.$$

Unit 4 Key Takeaway

Stability is settled by two signs: $f(c) = 0$ locates an equilibrium and the sign of $f'(c)$ (or, when that is zero, the sign of f on each side) classifies it. The logistic equation packages this with a stable carrying capacity K and fastest growth at $\frac{K}{2}$. Every Bernoulli equation yields to $v = y^{1-n}$, which trades the lone power of y for a clean linear solve.